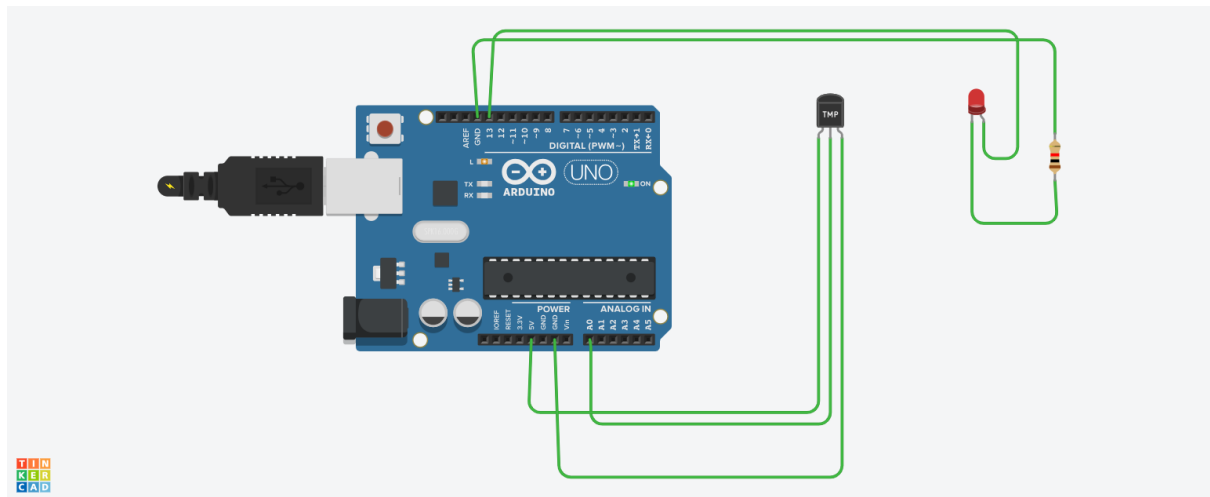


Embedded Systems & IOT – Task 2

NAME: KANNAN K

Embedded systems lab setup and sensor interfacing:

Project: Real Time Temperature Alert System using (TMP36 & Arduino)



```

1  const int SENSOR_PIN = A0;
2  const int LED_PIN = 13;
3
4  void setup() {
5      Serial.begin(9600);
6      pinMode(LED_PIN, OUTPUT);
7  }
8
9  void loop() {
10     int rawValue = analogRead(SENSOR_PIN);
11     float voltage = rawValue * (5.0 / 1023.0);
12     float tempC = (voltage - 0.5) * 100.0;
13
14     Serial.print("Temperature: ");
15     Serial.print(tempC);
16     Serial.println(" C");
17
18     if (tempC > 30.0) {
19         Serial.println("ALERT: High Temperature Detected!");
20         digitalWrite(LED_PIN, HIGH);
21     } else {
22         digitalWrite(LED_PIN, LOW);
23     }
24

```

 Serial Monitor

```

ALERT: High Temperature Detected!
Temperature: 72.19 C
ALERT: High Temperature Detected!
Temperature: 72.19 C
ALERT: High Temperature Detected!
Temperature: 72.19 C
ALERT: High Temperature Detected!

```

Source code: c++

```
const int SENSOR_PIN = A0;
```

```
const int LED_PIN = 13;
```

```
void setup() {
```

```
    Serial.begin(9600);
```

```
    pinMode(LED_PIN, OUTPUT);
```

```
}
```

```
void loop() {
```

```
    int rawValue = analogRead(SENSOR_PIN);
```

```
    float voltage = rawValue * (5.0 / 1023.0);
```

```
    float tempC = (voltage - 0.5) * 100.0;
```

```
    Serial.print("Temperature: ");
```

```
    Serial.print(tempC);
```

```
    Serial.println(" C");
```

```
    if (tempC > 30.0) {
```

```
        Serial.println("ALERT: High Temperature  
Detected!");
```

```
        digitalWrite(LED_PIN, HIGH);
```

```
    } else {
```

```
        digitalWrite(LED_PIN, LOW);
```

```
}
```

```
delay(1000);  
}
```

Project explanation:

Sensor Operation:

The TMP36 temperature sensor outputs an analog voltage proportional to ambient temperature.

Code Logic:

The Arduino reads the analog input on Pin A0 and converts the voltage reading into Celsius. The logic evaluates the reading every second. If the temperature exceeds 30°C , the system prints an alert message in the Serial Monitor and turns ON the LED connected to Pin 13.